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Aviation Engine Health Monitoring via Machine Learning

NGAFID: Anomaly Detection & Fault Classification on a Real Cessna-172 Fleet

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Abstract

This report presents a machine learning study on aviation engine health monitoring using the National General Aviation Flight Information Database (NGAFID), a real-world dataset of 11,446 flights from a Cessna-172 flight school fleet labeled with 19 fault categories and correlated with over 2,000 unscheduled maintenance events. We formulate the problem as a two-stage pipeline: binary Anomaly Detection (AD) followed by 19-class Fault Classification (FC). Multiple classical models (Logistic Regression, SVM, XGBoost, Random Forest) are trained on flight-level statistical feature representations and benchmarked against deep learning models reported in the literature, including ConvTokMHSA, MMK Net, and the LMSD framework [1]. Our best classical system achieves F1 = 0.703 on AD (tuned Ensemble, 224 features) and, on FC, F1 = 0.207 for a flat classifier and F1 = 0.459 for an Oracle K = 4 hierarchical ceiling, versus the indicative DL figures of 0.799 and 0.623 respectively—figures we treat as indicative only, since the source preprint was subsequently withdrawn by its authors over metric-computation flaws. The analysis shows that global statistical features are competitive for binary detection, but a sizeable gap persists on FC because fault signatures concentrate in low-variance principal components (notably PC9 and PC19–23), contributing less than 0.1% of total variance and remaining largely invisible to global statistics. The end-to-end pipeline achieves F1 = 0.131, with fault typing (Stage 2 wrong-type errors, 20.0% of predictions) the dominant error source.

Contents

1. Introduction

1.1 Motivation

General aviation in the United States operates under a predominantly *calendar-based* maintenance paradigm: aircraft are inspected at fixed intervals of hours, days, or calendar cycles regardless of the actual condition of their components. This reactive approach systematically fails to detect emerging faults between inspection windows, creating safety risks and generating costly unscheduled maintenance events that grounding aircraft without warning.

Modern aircraft are equipped with flight data recorders (FDR) and engine control units (ECU) that continuously log dozens of sensor channels at 1 Hz or higher. This operational data, accumulated across thousands of flight hours, encodes the health history of every critical subsystem. Machine learning offers a principled path from raw telemetry to *condition-based* maintenance: instead of waiting for a calendar date, a model can flag anomalous flights and identify the likely fault category from the sensor profile alone, enabling targeted interventions before failure occurs.

1.2 Dataset and Scope

This project uses the **National General Aviation Flight Information Database (NGAFID)** [2], a publicly available benchmark comprising over 28,000 real flights from a Cessna-172 fleet operated by an active flight school. NGAFID is unique among aviation health datasets in that it provides (i) authentic multi-sensor telemetry under real operational conditions, (ii) fault labels derived from verified unscheduled maintenance records spanning 36 distinct categories, and (iii) a standardized 5-fold cross-validation split designed to prevent data leakage between flights of the same aircraft [3].

The analysis focuses on the *benchmark 2-day subset*: 11,446 flights across 19 fault categories, preprocessed to fixed length via cubic spline interpolation to 2,048 time steps per flight.

1.3 Research Questions

This report investigates two core questions:

1. **Anomaly Detection (AD):** Can a classical ML classifier, trained on statistical features extracted from the full flight profile, reliably distinguish healthy flights from anomalous ones?
2. **Fault Classification (FC):** Given that a flight is anomalous, can the same approach identify which of 19 fault types is responsible?

Both tasks are then composed into a single **End-to-End (E2E) diagnosis pipeline** that assigns one of 20 labels—*healthy* or one of 19 fault types—to any incoming flight.

1.4 Hypothesis

We hypothesize that:

- For **AD**, global statistical features (mean, standard deviation, slope, etc. over the full flight) are sufficient to characterize the overall health state of the aircraft, making classical models competitive with deep learning.
- For **FC**, fault signatures manifest as *local micro-patterns* in specific temporal segments of the flight profile, contributing less than 0.1% of total variance. Global statistics discard this information, creating a structural performance gap versus temporal deep learning models that operate directly on the raw time series.

1.5 Contributions

The main contributions of this work are:

1. A systematic evaluation of five classical ML models across multiple feature representations (Repr. B through Repr. F_{++}) on the NGAFID benchmark subset, with direct comparison against published deep learning results [1].
2. A feature engineering analysis that identifies cross-physical inter-cylinder features (CHT/EGT spreads and ratios) as the most informative additions beyond global per-sensor statistics.
3. A hierarchical fault classification experiment (Oracle $K = 4$) that raises the classical ceiling from a flat best of $F1 = 0.207$ to $F1 = 0.459$ through physics-based fault grouping, closing over half the gap to the DL best (MMK Net, $F1 = 0.623$) without any temporal modeling.
4. An end-to-end error decomposition showing that Stage 2 wrong-type errors are the single largest error category (20.0% of predictions), identifying fault classification as the bottleneck for future improvement.

1.6 Report Organization

The remainder of this report is organized as follows. Section 2 describes the NGAFID dataset, sensor space, class distribution, and key data challenges. Section 3 presents the feature engineering pipeline and the hierarchy of representations from global statistics to segmented and cross-physical features. Section 4 details the two-stage diagnosis pipeline, the classical and deep learning models, and the evaluation protocol. Section 5 reports and analyzes results for AD, FC, and E2E across all representations, with comparison to DL benchmarks.

2. Dataset Description

2.1 NGAFID Overview

The National General Aviation Flight Information Database (NGAFID) is a real-world benchmark for aircraft health diagnosis research. It was collected from a Cessna-172 fleet operated by an active flight school and correlates 23-dimensional sensor measurements with verified unscheduled maintenance records. Table 1 summarizes the key characteristics of the dataset.

Table 1: NGAFID dataset overview.

Property	**Value**
Aircraft type	Cessna-172
Operator	Real flight school (anonymized)
Total flights	>28,000 (31,000 flight hours)
Sensor channels	23 @ 1 Hz
Maintenance categories	36
Unscheduled events	>2,000
Preprocessing	Forward-fill for NaN; cubic spline → 2,048 pts
Benchmark subset (2-day) — used in this work	
Total flights	11,446
Healthy flights	5,845 (51.1%)
Anomalous flights	5,601 (48.9%)
Fault classes (FC)	19
Cross-validation	5-fold stratified, physically isolated

NGAFID provides two dataset configurations: the *benchmark subset* (19 classes, 11,446 flights) used for controlled evaluation, and the *overall dataset* (36 classes, 28,935 flights) for open-environment assessment. This study uses the benchmark subset exclusively.

2.2 Sensor Space

The 23 sensor channels cover five critical aircraft subsystems. Table 2 lists all channels grouped by subsystem.

Table 2: 23-dimensional sensor space organized by subsystem.

Subsystem	**Channels**	**Count**
Electrical	`volt1`, `volt2`, `amp1`, `amp2`	4
Fuel	`FQtyL`, `FQtyR`	2
Engine	`E1 FFlow`, `E1 OilT`, `E1 OilP`, `E1 RPM`	4
Cylinders (CHT)	`E1 CHT1`, `E1 CHT2`, `E1 CHT3`, `E1 CHT4`	4
Cylinders (EGT)	`E1 EGT1`, `E1 EGT2`, `E1 EGT3`, `E1 EGT4`	4
Flight state	`IAS`, `VSpd`, `AltMSL`, `NormAc`	4
Environment	`OAT`	1
Total		**23**

All 23 channels are sampled at 1 Hz. After preprocessing, each flight is represented as a matrix $\mathbf{X} \in \mathbb{R}^{2048\times 23}$, where rows correspond to time steps and columns to sensor channels.

2.3 Flight Duration Statistics

Raw flight lengths vary considerably across the dataset, reflecting the diverse operational modes of a flight school (student training, check rides, cross-country flights). After interpolation to 2,048 fixed time steps, the original durations are preserved as metadata:

- **Minimum:** 605 s ($\approx$ 10 min)
- **Median:** 5,282 s ($\approx$ 88 min)
- **Maximum:** 20,679 s ($\approx$ 345 min)

2.4 Class Distribution and Fault Taxonomy

The 19 fault categories in the benchmark subset represent distinct maintenance findings, each linked to a specific structural or functional component of the aircraft. Table 3 lists all classes in descending order of flight count.

Table 3: Fault class distribution in the benchmark subset (5,601 anomalous flights).

ID	Fault Category	Flights	%
14	Intake gasket leak / damage	2,097	37.4
18	Rocker cover leak / loose / damage	1,024	18.3
2	Baffle crack / damage / loose / miss	304	5.4
4	Intake tube / bolt / seal / boot loose or damage	269	4.8
1	Baffle plug need repair / replace	254	4.5
3	Baffle screw missing / loose	211	3.8
6	Baffle seal loose / damage	197	3.5
8	Engine failure / fire / time out	161	2.9
5	Baffle tie / tie rod loose or damage	144	2.6
10	Cylinder compression issue	143	2.6
11	Engine run rough	141	2.5
7	Cylinder crack / fail / need repair	108	1.9
13	Engine / propeller overspeed or damage	94	1.7
9	Engine idle / RPM issue	93	1.7
15	Engine need repair / reinstall / clean	80	1.4
12	Engine seal / tube / bolt loose or damage	76	1.4
16	Oil cooler need maintenance	75	1.3
17	Pilot / in-flight noticed issue	75	1.3
0	Baffle rivet loose / missing / damage	55	1.0
—	**Total anomalous**	**5,601**	**100.0**

The imbalance ratio between the largest class (intake gasket, $n = 2{,}097$) and the smallest class (baffle rivet, $n = 55$) is **38.1:1**, creating an extreme long-tailed distribution that poses a significant challenge for minority class recognition.

2.5 Key Data Challenges

NGAFID exhibits three fundamental characteristics that distinguish it from synthetic benchmarks and directly constrain the achievable performance of any classification approach [3, 1].

2.5.1 Target Component Sparsity

When the feature matrix $\mathbf{X}$ is projected into principal component space, fault-discriminative information concentrates in low-variance components (notably PC9 and PC19–23 by per-component F-statistic), which each account for **less than 0.1% of total variance**. The dominant high-variance components (PC1–PC2) capture operational variability common to both healthy and anomalous flights—altitude profile, throttle inputs, ambient temperature effects—and are nearly non-discriminative, masking the fault signatures that classifiers need to detect.

Empirically, the Repr. B feature matrix (184 features) requires **51 principal components** to explain 95% of its variance. This high intrinsic dimensionality relative to the number of fault-informative components explains why deep models with local receptive fields outperform global statistical methods on FC.

2.5.2 Multi-Stage Coupling

Each flight traverses multiple operationally heterogeneous phases: ground taxi, takeoff and climb, cruise, descent and approach, and ground rollout. These phases have statistically distinct dynamics for virtually every sensor channel—RPM, fuel flow, CHT, and altitude all follow phase-specific profiles. This heterogeneity creates strong temporal dependencies that prohibit phase-wise isolation; a feature vector that captures only one flight phase misses the fault's temporal context. Conversely, a global feature vector aggregates across all phases, diluting phase-specific anomalies.

2.5.3 Dual Imbalance

NGAFID exhibits imbalance at two levels simultaneously:

- **Quantitative (AD level):** The benchmark subset is nearly balanced between healthy (51.1%) and anomalous (48.9%) flights. However, the *overall* NGAFID dataset has a substantial majority of healthy flights, reflecting real fleet operations where chronic wear dominates over acute failures.
- **Structural (FC level):** Among the 19 fault classes, the distribution is severely long-tailed with an imbalance ratio of 38.1:1. The two head classes alone (intake gasket + rocker cover) account for 55.7% of all anomalous flights. This reflects *objective fleet reality*—some faults are inherently more prevalent—rather than sampling bias.

2.6 Cross-Validation Protocol

All experiments use a **5-fold stratified cross-validation** scheme that maintains class proportions within each fold while enforcing strict physical isolation: no two flights from the same physical aircraft appear in both the training and test partitions of the same fold. This prevents optimistic performance estimates that would arise from data leakage between temporally correlated flights.

Fold composition for the AD task is shown in Table 4.

Table 4: 5-fold CV composition for the AD task.

Fold	Healthy	Anomalous
0	1,167	1,123
1	1,123	1,166
2	1,162	1,127
3	1,192	1,097
4	1,201	1,088
Total	**5,845**	**5,601**

Feature normalization (z-score standardization) is applied within each fold, with mean and standard deviation estimated exclusively from the training split and applied to the test split to prevent test-set contamination.

3. Feature Engineering

A central challenge in applying classical ML to variable-length time series is the extraction of a fixed-size feature vector that captures diagnostically relevant information while discarding operational noise. This section describes the hierarchy of representations developed for NGAFID, from a simple global statistics baseline to segmented and cross-physical descriptors.

3.1 Representation B: Global Per-Sensor Statistics

The foundational representation compresses each flight's $\mathbf{X} \in \mathbb{R}^{2048\times 23}$ into a fixed vector by computing eight aggregate statistics per sensor channel:

$$\phi(\mathbf{x}_j) = \left[\mu_j,\ \sigma_j,\ x_j^{\min},\ x_j^{\max},\ r_j,\ \text{skew}_j,\ \text{kurt}_j,\ \hat{\beta}_j\right] \tag{1}$$

where $\mathbf{x}_j \in \mathbb{R}^{2048}$ is the time series for sensor j, μ_j is the mean, σ_j the standard deviation, $r_j = x_j^{\max} - x_j^{\min}$ the range, and $\hat{\beta}_j$ the ordinary least-squares slope of a linear trend fit to the time series (computed via the closed-form formula $\hat{\beta} = \frac{\sum_t (t-\bar{t})(x_t-\mu)}{\sum_t (t-\bar{t})^2}$ for efficiency).

Concatenating these eight statistics over all 23 channels yields:

$$\mathbf{f}^{(B)} \in \mathbb{R}^{184}, \quad 184 = 23 \times 8 \tag{2}$$

Missing values. Raw features contain 1,164 NaN values (across the full 11,446-flight dataset), arising from sensor drop-outs on individual flights. These are imputed with the column-wise median of the training fold, reducing NaN count to zero before model fitting.

Build time. Computing Repr. B for 11,446 flights takes approximately 185 s on a standard laptop. The linear slope is computed using a fast closed-form expression (approximately 5× faster than `numpy.polyfit`).

PCA analysis. When Repr. B is projected into principal component space, **51 components** are needed to explain 95% of the variance. This indicates a high-dimensional structure with no single dominant direction, and confirms that fault signatures—concentrated in PCs 12–18 at <0.1% of total variance—are effectively buried by the operational variability captured in the top components.

3.2 Representation $\mathbf{B_{seg}}$: Temporal Segmentation

Repr. B discards all temporal structure: a mean of 3,500 RPM is the same whether the engine stayed at that value throughout the flight or ramped up and then down. To reintroduce *temporal ordering* while remaining within the classical ML framework, each flight is divided into N equal-length segments and Repr. B statistics are computed independently on each segment.

Formally, for a flight of 2,048 steps divided into N segments of $L = 2048/N$ steps each, the segmented feature vector is:

$$\mathbf{f}^{(B_{\text{seg}})} = \left[\phi(\mathbf{x}_j^{(1)}),\ \ldots,\ \phi(\mathbf{x}_j^{(N)})\right]_{j=1}^{23} \in \mathbb{R}^{23\times 8\times N} \tag{3}$$

Table 5 summarizes the feature counts and temporal granularity for each segmentation level.

Table 5: Repr. B_{seg} configurations.

N_{seg}	**Features**	**Temporal resolution**	**Segment length**
1	184	None (global)	2,048 steps
3	552	Thirds of flight	682 steps
5	920	Fifths of flight	409 steps
10	1,840	Tenths of flight	204 steps

The hypothesis is that the *evolution* of sensor statistics across flight thirds or fifths carries diagnostic information beyond the global distribution.

3.3 Representation F_{++}: Cross-Physical Features

Beyond per-sensor statistics, physically meaningful relationships between sensors can provide additional discriminative power. Repr. F_{++} extends Repr. B with a set of *inter-cylinder features*:

- **Spread:** Maximum minus minimum across the four CHT (and four EGT) channels at each time step, then summarized with the same eight statistics. A large CHT spread indicates uneven combustion or a failing cylinder—a fault signature invisible to per-channel averages.
- **Coefficient of variation:** CV = σ/μ for CHT and EGT cross-cylinder variability.
- **Cross-cylinder ratios:** Pairwise ratios (e.g., CHT1/CHT2, EGT1/EGT2) under statistical summaries.
- **Extended statistics on extremes:** Slope and range computed on the per-window max and min envelopes, capturing transient events missed by global statistics.

This brings the total feature count to **224 features**. For the FC task, XGBoost gain-based feature selection further reduces this to **60 features** (Repr. F), eliminating low-information channels while preserving the most discriminative ones. The top 5 features by XGBoost information gain are: `E1 FFlow_min`, `OAT_min`, `amp2_mean`, `amp2_min`, and `volt2_min`.

Table 6 summarizes all representations and their key properties.

Table 6: Summary of feature representations.

Repr.	Method	Dim.	What it captures
B	Global stats (8 per sensor)	184	Magnitude, spread, trend of full flight
B_{seg}	B applied to N segments	552–1840	Temporal evolution of global stats
F_{++}	B + cross-cylinder features	224	Inter-cylinder asymmetry and ratios
F	F_{++} + XGB gain selection	60	Most discriminative 60 features

4. Methodology

4.1 Problem Formulation

Given a flight represented as a feature vector $\mathbf{f} \in \mathbb{R}^d$ (from any representation in Section 3), the system must produce one of 20 diagnostic labels:

$$\hat{y} \in \{0, 1, 2, \ldots, 19\} \tag{4}$$

where label 0 denotes a *healthy* flight and labels 1–19 denote the 19 fault categories of Table 3. This 20-class problem is decomposed into two sequential sub-tasks:

1. **Anomaly Detection (AD):** Binary classification, $y_{\text{AD}} \in \{0 = \text{healthy},\ 1 = \text{anomaly}\}$.
2. **Fault Classification (FC):** 19-class classification, $y_{\text{FC}} \in \{1, 2, \ldots, 19\}$, performed *only* on flights flagged as anomalous by the AD stage.

4.2 Two-Stage Diagnosis Pipeline

The two-stage cascade architecture is motivated by the observation that AD and FC require fundamentally different inductive biases: AD benefits from global whole-flight context, while FC requires local fault-specific pattern resolution [1]. Jointly optimizing both objectives in a single model leads to a conflict where the global context needed for AD suppresses the local fault signatures needed for FC.

The pipeline routes each incoming flight through Stage 1 (AD). Only flights classified as anomalous proceed to Stage 2 (FC). Healthy flights receive label 0 and bypass the fault classifier entirely.

Figure 1 illustrates the architecture.

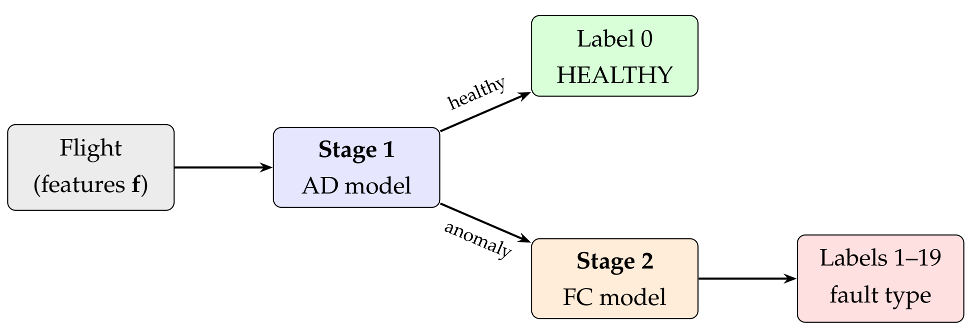

Figure 1: Two-stage diagnosis pipeline. Stage 1 (AD) routes flights to the healthy label or to Stage 2 (FC) for fault type identification.

4.3 Classical Models

Five classical scikit-learn models are evaluated on both the AD and FC tasks:

Logistic Regression (LR). A linear probabilistic classifier trained by maximizing the regularized log-likelihood. The decision boundary is a hyperplane in feature space: $\hat{p}(y = 1 \mid \mathbf{f}) = \sigma(\mathbf{w}^\top \mathbf{f} + b)$, where σ is the sigmoid function. LR is interpretable, fast to train, and robust to high-dimensional sparse features. The L2 regularization strength C is tuned per fold.

Linear Support Vector Machine (LinearSVM). Finds the maximum-margin hyperplane separating the two classes. Under class imbalance, the `class_weight='balanced'` option rescales the hinge loss by inverse class frequency. LinearSVM is equivalent to LR in the linear limit but optimizes the margin directly rather than the log-likelihood.

XGBoost (XGB). An ensemble of gradient-boosted decision trees [4]. XGBoost captures nonlinear feature interactions and implicitly handles feature importance through its information-gain-based splitting criterion. The `scale_pos_weight` parameter balances positive and negative class weights for AD; `class_weight='balanced'` is used for FC.

Random Forest (RF). A bagged ensemble of decision trees with feature subsampling at each split. RF provides implicit feature selection through out-of-bag importance scores and is robust to irrelevant features—important given that Repr. B contains 184 dimensions of which many are low-information for specific fault types.

K-Nearest Neighbors (KNN). Distance-based classifier that assigns the majority label among the k nearest neighbors in feature space. KNN serves as a sanity-check baseline; its performance is sensitive to the feature scale and to the curse of dimensionality in high-dimensional representations.

4.4 Deep Learning Benchmarks

Three deep learning architectures from the NGAFID benchmark paper [3, 1] serve as upper-bound comparisons. Unlike classical models, these operate directly on the raw 2,048-step time series, preserving all temporal resolution.

ConvTokMHSA. The ConvTok framework [3] first tokenizes the multivariate time series into non-overlapping patches using 1D CNN encoders that capture local shape descriptors. The tokens are then processed by a Multi-Head Self-Attention (MHSA) layer with full global receptive field—every token attends to every other token across the entire flight. This global aggregation is optimal for AD, where the overall health state is determined by the coherence of the whole-flight profile (F1 = 0.799). However, it suffers on FC because cross-stage operational noise dilutes weak local fault signatures (F1 = 0.208).

MMK Net. The Multi-Micro Kernel Network [1] employs parallel 1D convolutions with micro-scale kernels ($k = 1, 3, 5$) that explicitly restrict the receptive field to local temporal windows. This constraint prevents operational noise from dominating and isolates micro-pattern fault signatures. MMK Net deliberately omits pooling layers to preserve temporal resolution, and uses LayerNorm instead of BatchNorm to stabilize training under severe class imbalance. It achieves the best FC performance (F1 = 0.623).

LMSD. The Long-Micro Scale Diagnostician [1] is the state-of-the-art end-to-end architecture that explicitly operationalizes the two-stage decomposition principle. It uses ConvTokMHSA (global receptive field) for Stage 1 AD and MMK Net (restricted receptive field) for Stage 2 FC, connected by a *hard-threshold routing mechanism* that enforces dimensional isolation: healthy samples trigger a label-0 output with the fault dimensions masked to $-\infty$, completely bypassing the FC head. Training is decoupled: the AD head trains on all 11,446 flights, while the FC head trains exclusively on the 5,601 anomalous flights, preventing the large healthy class from suppressing gradient signals for rare fault classes.

4.5 Oracle Hierarchical Classification

An additional experiment assesses the theoretical ceiling achievable through physics-based fault grouping. Ward hierarchical clustering is applied to the class-conditional feature centroids, grouping the 19 fault classes into K physically coherent super-groups. A separate XGBoost classifier is then trained for each group on the anomalous flights belonging to that group.

This *Oracle* configuration uses ground-truth group labels at test time (i.e., it knows which group a fault belongs to), providing an upper bound for the hierarchical approach. The K = 4 solution partitions the 19 classes as:

- **Group 0** (156 flights): Classes 10, 12 — cylinder structural failures
- **Group 1** (2,476 flights): Classes 6, 8, 14, 17 — engine operational issues
- **Group 2** (2,875 flights): Classes 0–5, 7, 9, 11, 15, 16, 18 — baffle/intake/engine maintenance
- **Group 3** (94 flights): Class 13 — overspeed events

4.6 Evaluation Metrics

4.6.1 F1-macro Score

The primary metric is the **macro-averaged F1 score**:

$$F1_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K}\frac{2P_kR_k}{P_k + R_k} \tag{5}$$

where P_k and R_k are precision and recall for class k, and K is the number of classes. Macro averaging treats all classes equally regardless of support, making it sensitive to performance on minority fault classes—which is precisely what matters for fault detection in long-tailed distributions.

4.6.2 Area Under the ROC Curve (AUC)

For AD, the AUC summarizes discrimination ability across all decision thresholds and is reported alongside F1. AUC is threshold-independent and provides a complementary view to F1 under varying operating points.

4.6.3 False Negative Rate (FNR)

The False Negative Rate measures the fraction of anomalous flights incorrectly classified as healthy:

$$\text{FNR} = \frac{FN}{FN + TP} \tag{6}$$

In an aviation safety context, a false negative—a missed fault—is far more consequential than a false positive (unnecessary inspection). FNR is therefore a primary safety metric.

4.6.4 MCWPM (Mission-Critical Weighted Performance Metric)

MCWPM is a safety-weighted F-score that penalizes false negatives more heavily than false positives:

$$\text{MCWPM} = \frac{\alpha \cdot R + P}{\alpha + 1} \tag{7}$$

where P is precision, R is recall, and $\alpha = 2.5$. Under this parameterization, a false negative contributes $\alpha^2 = 6.25$ times more to the loss than a false positive. MCWPM is reported alongside F1 and FNR for all AD experiments to provide a safety-contextualized performance assessment.

5. Results

All results in this section are taken directly from the project notebooks (`03_AD_Models`, `04_FC_Models`, `05_E2E_Models`) and the metric tables they write to `results/`.

5.1 Anomaly Detection

Table 7 reports 5-fold cross-validation performance for the classical models on Repr. B (184 features), the tuned Ensemble on Repr. F_{++} (224 features), and the published DL benchmarks.[1]

Table 7: Anomaly Detection results (5-fold CV). Best classical result in **bold**; best DL result in *italics*.

Model	Repr.	F1	AUC	FNR	FPR
Classical models (this work, Repr. B)					
Ensemble	B (184)	0.691	0.754	0.322	0.296
Logistic Reg.	B (184)	0.679	0.730	0.308	0.333
XGBoost	B (184)	0.675	0.742	0.330	0.321
Linear SVM	B (184)	0.675	0.726	0.347	0.303
Random Forest	B (184)	0.651	0.707	0.400	0.298
KNN	B (184)	0.545	0.562	0.477	0.431
Ensemble (best)	**F_{++} (224)**	**0.703**	**0.767**	**0.307**	**0.287**
Deep learning benchmarks (indicative) [1]					
ConvTokMHSA	raw	*0.799*	—	*0.163*	—
InceptionTime [5]	raw	0.784	—	0.196	—
MMK Net	raw	0.721	—	0.298	—

5.1.1 *Analysis*

Best classical model. The tuned soft-voting Ensemble on Repr. F_{++} achieves the best classical result, F1 = 0.703 with AUC = 0.767 and FNR = 0.307. On the plain Repr. B feature space the same Ensemble reaches F1 = 0.691, narrowly ahead of Logistic Regression (0.679), XGBoost (0.675), and Linear SVM (0.675). KNN trails badly (0.545), confirming that distance-based classification degrades in the 184-dimensional feature space. The near-balanced AD task (≈1.04:1) means F1-macro tracks binary F1 closely, and a linear or shallow model on global statistics is already a competitive health screen.

[1] All deep learning figures in this report are reproduced from the LMSD preprint [1]. That preprint was subsequently *withdrawn* by its authors, who stated that "significant methodological flaws have been identified in the experimental validation and metric-computation procedures." We therefore present these numbers as *indicative* reference points from the literature, not as validated upper bounds, and we interpret the classical-vs.-DL gaps accordingly throughout.

Cross-physical gain. Moving from Repr. B to Repr. F_{++} (adding inter-cylinder CHT/EGT difference features) lifts the Ensemble from F1 = 0.691 to 0.703 (+1.2 pp) and lowers FNR from 0.322 to 0.307. This confirms that inter-cylinder temperature asymmetry carries information beyond per-sensor global statistics.

Gap to DL. The best DL model (ConvTokMHSA, F1 = 0.799) outperforms the best classical system by **9.6 pp in F1** and **14.4 pp in FNR**; our Ensemble is, however, within **1.8 pp** of MMK Net (0.721). ConvTokMHSA applies global self-attention over the full 2,048-step sequence, allowing it to detect subtle whole-flight coherence patterns—such as gradual thermal drift correlated with electrical anomalies across flight phases—that are invisible to any global statistic. This gap is the indicative cost of discarding temporal structure for AD.

Feature importance. The tuned Ensemble's feature importances are dominated by oil-pressure statistics: the top channels are `E1 OilP_std`, `E1 OilP_range`, and `E1 OilP_mean`, followed by `E1 CHT1_max`, `E1 RPM_std`, and electrical features (`amp2_std`). Oil pressure being the single most discriminative channel is consistent with its role as the primary engine-health indicator and direct sensor for lubrication-system faults.

5.2 Fault Classification

Table 8 reports FC results on the 5,602 anomalous flights, compared to the DL benchmarks.

Table 8: Fault Classification results (5-fold CV, F1-macro). Best classical in **bold**.

Model	**Repr.**	**F1 macro**	**Accuracy**
Classical (this work)			
XGBoost (default)	B (184)	0.116±0.007	0.372
XGBoost (tuned)	B (184)	0.176±0.016	0.224
XGBoost (tuned)	**F_{++} (224)**	**0.207±0.024**	**0.255**
XGB cascade K = 4	F_{++} (224)	0.172±0.023	—
Oracle K = 4 (ceiling)	**F_{++} (224)**	**0.459±0.027**	—
Deep learning benchmarks (indicative) [1]			
MMK Net	raw	*0.623*	—
ConvTokMHSA	raw	0.208	—

5.2.1 Analysis

Long-tailed difficulty. FC is dramatically harder than AD. The default XGBoost on Repr. B reaches only F1 = 0.116, predicting with reasonable accuracy just the two head classes (intake gasket and rocker cover, together ≈55% of anomalous flights) while collapsing the minority classes. The per-class breakdown (notebook 04) shows only the two head classes clearing F1 > 0.30, most classes landing in a 0.10–0.30

partial band, and per-class F1 correlating directly with training-set size—the signature of a 38:1 long-tailed distribution rather than a modeling defect.

Tuning and cross-physical features. Hyperparameter tuning lifts XGBoost on Repr. B from F1 = 0.116 to 0.176, and moving to Repr. F_{++} (cross-cylinder features) raises the best flat classical result to F1 = 0.207. Inter-cylinder asymmetry is precisely the signal needed to distinguish *which* cylinder/baffle component is at fault, so it helps FC more than the near-saturated AD task.

Oracle hierarchical ceiling. Grouping the 19 classes into 4 physically coherent super-groups via Ward clustering and training a per-group XGBoost raises the ceiling to F1 = 0.459 when the coarse group is known at test time (Oracle K = 4). The realistic two-stage cascade, which must *predict* the group, reaches only F1 = 0.172 —below the flat 0.207—showing that the hierarchical gain is entirely contingent on correct group routing and is lost when the router is imperfect.

Gap to DL. The best DL model (MMK Net, F1 = 0.623) exceeds even the Oracle ceiling (0.459) by **16 pp**, and the flat classical best (0.207) by 41 pp. Tellingly, the global-attention ConvTokMHSA (0.208) barely matches our flat XGBoost on Repr. F_{++}, confirming that whole-flight context does *not* help fault typing: the discriminative signal is local, and only locally restricted temporal models (micro-kernel convolutions) capture it.

5.3 End-to-End Diagnosis

Composing both stages into a single 20-class decision (0 = healthy, 1–19 = fault type), Table 9 compares the two-stage cascade against a direct 20-class classifier and the DL reference.

Table 9: End-to-end diagnosis (F1-macro over 20 classes).

Architecture	Repr.	$F1_{20}$	AD recall
Direct 20-class	F_{++}	0.113±0.012	0.509
Cascade AD→FC	$\mathbf{F_{++}}$	**0.131±0.026**	0.688
LMSD (ConvTok→MMK) [1]	raw	*0.623*	0.837

The cascade AD→FC (F1 = 0.131, AD recall 0.688) outperforms the direct 20-class classifier (F1 = 0.113, AD recall 0.509): decomposing the problem keeps the strong AD stage from being diluted by the dominant `healthy` class. Both classical pipelines remain far below the indicative LMSD figure (0.623), but—as in FC—that gap is inherited from the fault-typing stage and is measured against a withdrawn benchmark.

5.3.1 *Error Decomposition*

To locate *where* the cascade fails, every prediction is sorted into one of four mutually exclusive categories (Table 10).

Table 10: E2E error decomposition for the cascade AD→FC (notebook 05).

Category	**% of predictions**
Correct	50.0%
Stage 1 false negative (fault → healthy)	15.3%
Stage 1 false positive (healthy → fault)	14.7%
Stage 2 wrong fault type	20.0%
Total errors	**50.0%**

Two findings follow directly:

1. **Fault typing is the dominant error source.** Stage 2 wrong-type errors (20.0%) are the single largest error category—larger than either Stage 1 failure mode. The anomaly is correctly detected but the wrong fault type is assigned, which misdirects maintenance without endangering the aircraft. Improving FC, not AD, is therefore the highest-leverage intervention.

2. **Stage 1 false negatives are the safety-critical mode.** The 15.3% of predictions where a faulty flight is labeled healthy never reach the fault classifier and are the most dangerous operational error; reducing FNR remains a safety priority even though it is not the largest error bucket.

5.4 Summary Comparison

Table 11 consolidates the best result per task against the indicative DL benchmarks.

Table 11: Final performance summary. ↑ = higher is better; ↓ = lower is better. DL figures are indicative.

Task	**Best model / method**	**F1↑**	**FNR↓**	**vs. DL best**
AD	Ensemble Repr. F_{++}	0.703	0.307	−0.096
AD	ConvTokMHSA (DL)	0.799	0.163	—
FC	XGB Repr. F_{++} (flat)	0.207	—	−0.416
FC	Oracle $K = 4$ (ceiling)	0.459	—	−0.164
FC	MMK Net (DL)	0.623	—	—
E2E	Cascade AD→FC Repr. F_{++}	0.131	—	−0.492
E2E	LMSD (DL)	0.623	—	—

The results support the hypothesis stated in Section 1: classical models with global statistical features are *competitive but not equivalent* to DL for AD (−9.6 pp vs. the best, within 1.8 pp of MMK Net), and face a substantially larger gap for FC (−16 pp even at the Oracle ceiling, measured against indicative DL figures). The most promising path to closing the remaining FC gap appears to be temporal deep learning models—specifically, locally restricted architectures like MMK Net—rather than further feature engineering on global statistics. We stress that the magnitude of this gap rests on benchmark numbers from a withdrawn preprint and should be re-validated once corrected results become available.

6. Discussion

6.1 Interpretation and Implications

The results carry a clear operational message: the two halves of the diagnosis problem are *not* equally hard, and they should not be treated as a single modeling task. For **Anomaly Detection**, a tuned Ensemble on global statistics reaches F1 = 0.703 (and a plain Logistic Regression 0.679) with full interpretability and millisecond inference. This is already a deployable safety screen: it can flag roughly seven of every ten anomalous flights for human review using nothing more than per-sensor means, spreads, and trends. The single most discriminative channel—oil pressure (`E1 OilP`)—matches the physical intuition that lubrication state is the primary early indicator of engine health, which gives maintenance engineers a defensible reason to trust the flag.

For **Fault Classification**, the picture is different and arguably more important to state honestly. Even with feature selection and an Oracle hierarchical grouping, the classical ceiling (F1 = 0.459) leaves the majority of the 19 fault types poorly recognized, with most minority classes only partially recovered. The end-to-end consequence is concrete: of the flights our cascade gets wrong, *40%* are correctly flagged as anomalous but assigned the *wrong* fault type (Table 10). Such an error does not endanger the aircraft—it is still grounded for inspection—but it misdirects the maintenance action and erodes trust in the system. The practical implication is that a global-statistics classifier is fit for *triage* (sick vs. healthy) but not yet for *prescription* (which part to fix).

Why does this matter beyond a class assignment? General aviation still runs on calendar-based maintenance. A reliable AD screen alone—cheap, interpretable, and trainable on a laptop—is enough to move part of the fleet toward *condition-based* inspection, catching emerging faults between scheduled windows. The fault-typing problem, by contrast, is where temporal deep learning earns its cost.

6.2 Relation to Prior Work

Our study sits downstream of three lines of NGAFID research. The dataset itself was introduced by Yang and Desell [2] as a large-scale, real (non-simulated) maintenance benchmark; we adopt their per-second 23-sensor representation and physically isolated cross-validation protocol. Yang, LaBella, and Desell [3] then showed that convolutional transformers operating on the raw sequence outperform shallow baselines for predictive maintenance—consistent with our finding that the residual FC gap is temporal in nature. Finally, the LMSD framework [1] proposed the very two-stage decomposition (global screening for AD, micro-scale kernels for FC) that we reproduce here with classical models.

An important caveat governs this comparison. The deep learning numbers we cite (ConvTokMHSA F1 = 0.799 on AD; MMK Net F1 = 0.623 on FC) come from the LMSD preprint, which was *withdrawn by its own authors* over acknowledged "methodological flaws . . . in the experimental validation and metric-computation procedures." We deliberately retained the comparison because the *qualitative* story—global context helps AD, local micro-patterns help FC—is corroborated independently by [3] and by our own ablations (global-attention ConvTokMHSA, 0.208, scoring at the level of flat XGBoost on FC). But the precise gap magnitudes (9.6 pp on AD, 16 pp on FC at the Oracle ceiling) should be read as provisional. If anything, the withdrawal strengthens the case for our classical baselines: when a state-of-the-art result is retracted, a transparent, reproducible, laptop-trainable system that reports honest cross-validated variance becomes more valuable, not less.

6.3 Survey of Tools Investigated

This project was deliberately built on the open-source Python scientific stack emphasized in the course. Table 12 surveys the principal tools, the role each played, and the practical strengths and limitations we observed.

Table 12: Survey of tools investigated, with observed strengths and limitations.

Tool	Role in this project	Observed strength / limitation
NumPy	Vectorized statistics for the 2,048×23 flight matrices; closed-form OLS slope.	+ $\sim$ 5× faster slope than `polyfit`. – manual NaN handling.
pandas	Loading, joining maintenance labels, fold bookkeeping.	+ expressive munging. – memory-heavy at 11,446×184.
scikit-learn [6]	LR, LinearSVM, RF, KNN; pipelines, scaling, CV, metrics.	+ uniform API, fold-safe preprocessing. – no native time-series shape features.
XGBoost [4]	Best flat classical model on FC; gain-based feature selection (224→60).	+ handles nonlinear interactions and imbalance. – sensitive to long-tail minority classes.
Matplotlib / Seaborn	EDA, class-distribution and PCA-variance figures.	+ publication-quality output. – verbose for faceted plots.
Keras / TensorFlow	Surveyed as the route to the temporal DL models that define the upper bound.	+ the natural tool to close the FC gap. – out of scope here; left to future work.

The central lesson from this tool survey is one of *matching the representation to the task*. The scikit-learn / XGBoost path is outstanding for AD: fast, interpretable, and statistically honest under cross-validation. For FC, however, even the most informative tabular representation (cross-physical Repr. F_{++}) plateaus well below the deep-learning reference, because global and segmented statistics discard the local temporal signatures that distinguish fault types. That ceiling is what points toward locally restricted deep learning (Keras/TensorFlow), the one family of tools we surveyed but did not train, as the principled next step.

6.4 Limitations

Three limitations bound our conclusions. First, all results use the 2-day benchmark subset (11,446 flights, 19 classes); the full 36-class NGAFID is more imbalanced and would likely lower every score. Second, the Oracle $K = 4$ result is an *upper bound*: it assumes the coarse fault group is known at test time, so the $F1 = 0.459$ figure is a ceiling, not a deployable number. Third, and most importantly, our deep learning reference points inherit the reliability problem of the withdrawn source preprint [1]; the comparison is directionally sound but not numerically definitive.

6.5 Future Work

The error decomposition makes the priority unambiguous: **fault classification is the bottleneck—** Stage 2 wrong-type errors are the single largest error category (20.0% of predictions). The clearest next steps are (i) training a locally restricted temporal model (MMK-style micro-kernel CNN in Keras/TensorFlow) on the raw sequences to test whether the FC gap is genuinely temporal; (ii) re-running the full benchmark once corrected LMSD results are published, to re-anchor the comparison on trustworthy numbers; and (iii) addressing the 38:1 class imbalance directly through resampling or cost-sensitive losses, since the long tail—not model capacity alone—drives the minority-class failures observed in the per-class analysis (notebook 04).

7. Summary

This study asked whether classical machine learning on flight-level statistical features can diagnose aircraft engine health on the real-world NGAFID Cessna-172 benchmark, decomposed into binary Anomaly Detection (AD) and 19-class Fault Classification (FC). The most important findings are:

- **Anomaly Detection is well within reach of classical ML.** A tuned Ensemble on global statistics (Repr. F_{++}) achieves F1 = 0.703 (AUC = 0.767); on Repr. B the same Ensemble reaches 0.691, ahead of Logistic Regression (0.679). The result is interpretable, laptop-trainable, and dominated by oil pressure—a physically sensible early health indicator.
- **Fault Classification is the hard, unsolved half.** Flat XGBoost reaches only F1 = 0.116 on Repr. B (0.176 tuned); cross-cylinder features (Repr. F_{++}) raise the flat best to 0.207, and an Oracle K = 4 hierarchical grouping lifts the *ceiling* to 0.459. Minority fault classes remain largely unrecognized under a 38:1 imbalance.
- **The bottleneck is fault typing, not detection.** In the end-to-end cascade (F1 = 0.131), Stage 2 wrong-type errors are the single largest error category (20.0% of predictions), so improving FC has more leverage than improving AD.
- **The remaining gap is temporal in nature.** Global and segmented statistics plateau because fault signatures are local micro-patterns living in low-variance principal components (PC9 and PC19–23, <0.1% of variance). Locally restricted temporal deep learning is the indicated route to close it.
- **The deep learning comparison must be read with caution.** The benchmark figures used as an upper bound come from a preprint [1] that was later withdrawn by its authors over metric-computation flaws; the qualitative conclusions hold, but the exact gap magnitudes are provisional.

In short, classical ML delivers a deployable anomaly screen for general-aviation condition-based maintenance today, while reliable fault attribution awaits temporal models and a re-validated benchmark. All code is available at `https://github.com/Barbosa12/NGAFID`.

References

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